1. We will use mathematical induction to prove the following statement is true for all integers n greater than 7.

$$3^n < n!$$

We proceed by induction to prove P(n) is true for n ≥ 7, where P(n) = "$3^n < n!$ ".

For the base case, we have n = 7. By simple calculation of the left-hand side of the equation, we have $3^7 = 2187$. By simple calculation of the right-hand side of the equation, we have 7! = 5040. Therefore, we have found that for n = 7, we have $3^n < n!$.

For the inductive hypothesis, we will assume P(k) is true for some k ≥ 7.

For the inductive step, we wish to show that $3^{k+1} < (k+1)!$. We will take note that $3^{k+1} = 3^k(3) < (3)$k! where the inequality follows by the inductive hypothesis. Furthermore, for k ≥ 7, we know k + 1 ≥ 8 > 3. Thus, we have $(2)k! < (8)k! \leq (k+1)k! = (k+1)!$. We will combine these two equations to produce $3^{k+1} < 3k! < (k+1)!$ as desired. Therefore, we have finished our proof.

2. We will use mathematical induction to prove that for all positive integers n:

$$3|(n^3 + 2\text{n})$$

We proceed by induction to prove P(n) is true for all positive integers n, where
P(n) = "$3|(n^3 + 2\text{n})$".

For the base case, we have n = 1. By simple calculation of the right-hand side of the equation, we have $1^3 + 2(1)$, which simplifies to 3. Therefore, the resulting equation is 3|3, which is found to be true since 3 is a multiple of 3, therefore, 3 divides 3.

For the inductive hypothesis, we will assume 3 divides $k^3 + 2\text{k}$ for some integer k ≥ 1.

For the inductive step, we wish to show 3 divides $(k + 1)^3 + 2(\text{k}+1)$ or, in other words, $(k + 1)^3 + 2(\text{k}+1)$ is a multiple of 3. We will expand the above equation to produce $k^3 + 3k^2 + 5k + 3$. By the inductive hypothesis, we find there exists a positive integer q such that $k^3 + 2k = 3\text{q}$. We will substitute this into the previous equation to produce $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 5k + 3 = k^3 + 3k^2 + 2k + 3k + 3 = 3q + 3k^2 + 3$ which, by factoring out a three, produces $3(\text{q} + k^2 + 3)$. Since integers are closed under addition, $\text{q} + k^2 + 3$ is an element of integers, and we conclude that $(k + 1)^3 + 2(\text{k}+1)$ is a multiple of three. Therefore, our proof is finished.

3. We will use strong induction to find a valid smallest value x such that you can form postage of x cents and greater using only 5 cent and 8 cent stamps.

Using the Chicken McNugget Theorem, we find the smallest value x such that the above condition holds true is $x = mn - m - n + 1 = (5)(8) - 5 - 8 + 1 = 27 + 1 = 28$.

We will proceed by strong induction to prove P(x), where P(x) = "you can form postage of x cents and greater using only 5 cent and 8 cent stamps".

For the base case, we have the following:

$x = 28 \quad = 5(4) + 8(1)$

$x = 29 \quad = 5(1) + 8(3)$

$x = 30 \quad = 5(6) + 8(0)$

$x = 31 \quad = 5(3) + 8(2)$

$x = 32 \quad = 5(0) + 8(4)$

For the inductive hypothesis, we will assume
$P(28) \wedge P(29) \wedge P(30) \wedge P(31) \wedge P(32) \wedge \ldots \wedge P(k)$ are true for k.

For the inductive step, we will make change for $k + 1$. By the inductive hypothesis, we can make change for $k - 4 = (k + 1) - 5$. Then, by adding one 5 cent stamp, we get change for $k + 1$. Therefore, we have concluded our proof.

4. We will use strong induction to prove that $a_n \leq 2^n$ for all n ≥ 2, where $a_0 = 1, a_1 = 3, a_t = a_{t-1} + a_{t-2}$. We have P(n) = "$a_n \leq 2^n$".

For the base case, we have the following:

n = 2 $\quad a_2 = 3 + 1 = 4 \leq 2^2 = 4$

n = 3 $\quad a_3 = 4 + 3 = 7 \leq 2^3 = 8$

For the inductive hypothesis, we assume P(2) $\wedge P(3) \wedge \ldots \wedge P(k)$ are true for k ≥ 2.

For the inductive step, we will notice $a_{k+1} = a_t = a_k + a_{k-1} = a_{t-1} + a_{t-2}$. By the inductive hypothesis, we see $a_k \leq 2^k$and $a_{k-1} \leq 2^{k-1}$. By adding on both sides, we have $a_{k+1} = a_k + a_{k-1} \leq 2^k + 2^{k-1} < 2^k + 2^k = 2(2^k) = 2^{k+1}$. Finally, we see $a_{k+1} \leq 2^{k+1}$. We see that $P(k+1)$ is true whenever the inductive hypothesis is true. This therefore completes the inductive step. Therefore, we have proven what was desired for all n ≥ 2.